JIAER RUBBER & PLASTIC CO. | 佳尔橡塑 • JIAERXIANGSU

MT51-WS-1

TECHNICAL DATA SHEET

High-Performance Weather-Resistant & Water-Sealing Elastomer

Engineered Polymer Formulation for Industrial Sealing & Insulation

DOCUMENT NO.	JE-TDS-MT51-WS-1
REVISION	Rev 2026.10
QUALITY GRADE	Industrial / OEM
STATUS	Certified Release

01 EXECUTIVE PRODUCT OVERVIEW & COMMERCIAL ADVANTAGE

Material Family: Cross-Linkable Elastomer (WS-1)

JIAER MT51-WS-1 is an advanced, commercial-grade polymer and elastomer formulation developed by Jiaer Rubber & Plastic (佳尔橡塑 • JIAERXIANGSU) for demanding industrial sealing, waterproofing, and structural insulation applications. Engineered to maximize operational uptime and reduce total lifecycle maintenance costs, MT51-WS-1 combines high tensile resilience, hydrophobic weather sealing, and superior thermal-oxidative stability.

- **Maximum ROI & Longevity:** Extended service life under continuous compression & UV exposure.
- **Process & OEM Efficiency:** Consistent rheology & stability for high-yield extrusion/molding.
- **Zero-Leakage Reliability:** Hydrophobic WS-1 barrier preventing water and vapor ingress.

TENSILE STRENGTH	ELONGATION AT BREAK	SERVICE TEMPERATURE	WATER RESISTANCE
⩾ 14.5 MPa	⩾ 420 %	-40°C to +125°C	WS-1 Grade
ASTM D412 / ISO 37	High Elastic Recovery	Continuous Thermal Rating	⩽ 0.8% Absorption (24h)

02 TECHNICAL SPECIFICATIONS & PHYSICAL PROPERTIES

Standard Test Conditions: 23°C ± 2°C, 50% ± 5% RH

Property / Parameter	Test Method / Standard	Unit	MT51-WS-1 Specification
Material Classification	JIAER Internal / ISO 1629	—	Modified Weather-Sealing Elastomer (WS-1)
Appearance & Color	Visual Inspection	—	Uniform Smooth Surface / Black or Custom OEM
Specific Gravity / Density	ASTM D297 / ISO 2781	g/cm³	1.18 ± 0.03
Hardness (Shore A)	ASTM D2240 / ISO 7619-1	Shore A	60 ± 5 (Customizable Range: 50–75)
Tensile Strength (Min.)	ASTM D412 / ISO 37	MPa	⩾ 14.5
Elongation at Break (Min.)	ASTM D412 / ISO 37	%	⩾ 420
Tear Resistance (Die C)	ASTM D624	kN/m	⩾ 38.0
Compression Set (70°C × 22h)	ASTM D395 Method B	%	⩽ 22
Water Absorption (23°C × 24h)	ASTM D471	%	⩽ 0.8 (Hydrophobic WS-1 Barrier Grade)
Ozone & Weathering Resistance	ASTM D1149 (50 pphm, 72h)	Rating	No Cracks / Pass Grade A
Thermal Air Aging (100°C × 70h)	ASTM D573	Δ Value	Hardness: ⩽ +5 Shore A | Tensile: ⩽ -10%

03 APPLICATIONS, PROCESSING GUIDELINES & QUALITY ASSURANCE

ISO 9001 / RoHS / REACH Compliant

TARGET INDUSTRIAL APPLICATIONS

- Automotive & Commercial Weatherstripping
- Architectural Curtain Wall & Glazing Seals
- Waterproof Conduit, Cable & Pipe Gaskets
- HVAC Vibration Damping & Insulation
- Heavy Machinery Environmental Profiles

PROCESSING, PACKAGING & STORAGE

- Process: Extrusion, Compression & Injection
- Packaging: 25 kg barrier bags / 500 kg pallet
- Shelf Life: 12 months in dry storage (<30°C)
- Compliance: RoHS, REACH & ISO 9001 Certified
- Custom OEM: Hardness, color & cure profiles

AUTHENTIC OEM MATERIAL
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Disclaimer: Typical technical values are provided for engineering reference and do not constitute a binding specification without OEM batch certification.